

Shopify Custom Door Configurator – System Documentation

Overview

This document explains how the product configuration system has been built and how your team can easily manage updates in the future without modifying theme code.

The entire system was designed with a **dynamic architecture** so that product options, pricing rules, category assignments, and product tab content can be managed directly from the **Shopify Admin dashboard**.

No data has been hard-coded into theme files. Instead, the system uses **Shopify Metafields** so everything can be edited without requiring a developer.

1. System Architecture

The system uses the following components:

Shopify Metafields

Used to store:

- Product configuration options
- Pricing rules
- Product categories
- Product tab content

Theme Logic (Liquid + JavaScript)

The storefront theme dynamically:

- Reads metafield data
- Builds the configurator interface
- Calculates pricing
- Displays product details

Because the theme only **reads data from metafields**, any update in Shopify Admin immediately updates the storefront.

2. Custom Door Options (Door Configurator Data)

The **custom.door_options** metafield stores the configuration data used by the Door Configurator.

This field contains **JSON formatted data** that defines all configurable product options.

These options may include:

- Size (*imported from the previous Magento system*)
- Door Configuration
- Pre Hanging
- Jamb Species
- Handing
- Jamb Width
- Hinges Finish
- Handle Preparation

Important Note About Size

The **Size** option may appear in the JSON data because it was imported from the Magento system.

However, **the actual size selection on the storefront is controlled by Shopify Product Variants**, not by the configurator logic.

This means:

- Size in the JSON data **does not affect the storefront behavior**
- The storefront uses **Shopify variants** for size selection
- Only **Door Configuration and other door options** need to be modified in the JSON data when changes are required

Therefore, even if the **Size object remains in the JSON structure**, it does **not impact the configurator functionality**.

JSON Data Structure

The metafield stores a **JSON array**, where each object represents a configurable option.

Example structure:

```
[
  {
    "id": "24858",
    "title": "Size",
    "type": "radio",
    "is_require": true,
    "sort_order": 1,
    "values": [
      {
        "value_id": "228261",
        "title": "15\\x80\\\"",
        "price": 289,
        "price_type": "fixed",
        "sort_order": 1
      }
    ]
  },
  {
    "id": "24859",
    "title": "Door Configuration",
    "type": "radio",
```

```
"is_require": false,
"sort_order": 2,
"values": [
  {
    "value_id": "228317",
    "title": "Single Door",
    "price": 0,
    "price_type": "fixed",
    "sort_order": 1
  },
  {
    "value_id": "228318",
    "title": "Double Door",
    "price": 0,
    "price_type": "fixed",
    "sort_order": 2
  }
]
}
```

Field Explanation

Option Level Fields

Field	Description
id	Unique identifier for the option
title	Name of the option displayed on the storefront
type	Input type (radio, dropdown, field etc.)
is_require	Determines if the option must be selected
sort_order	Controls the order in which options appear
values	List of selectable values

Option Value Fields

Field	Description
value_id	Unique identifier for the option value
title	Label displayed to the customer
price	Price adjustment applied when selected

price_type Pricing rule type (fixed price adjustment)

sort_order Order of the values within the option

How the Configurator Uses This Data

When the product page loads:

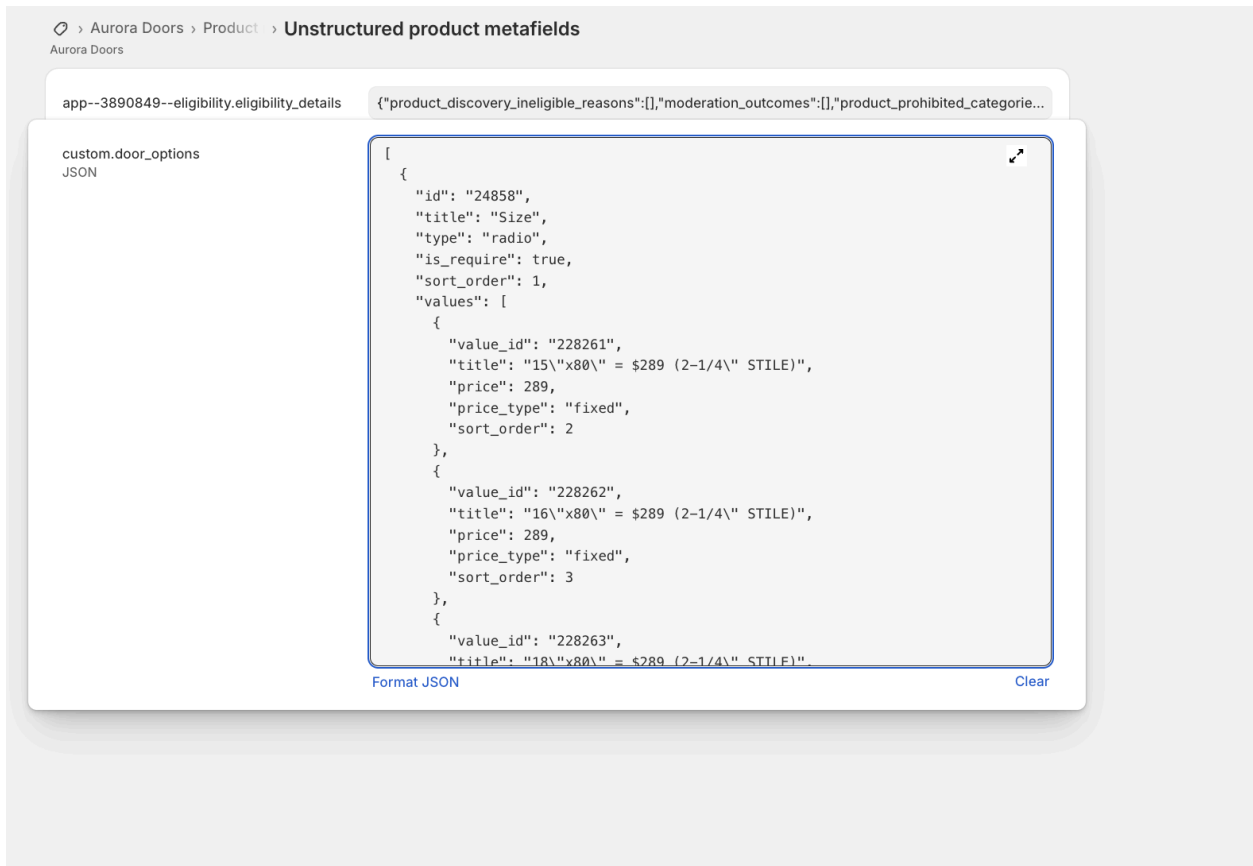
1. The storefront reads the **custom.door_options** metafield.
 2. The JSON data is parsed by the configurator script.
 3. The configurator dynamically generates the option interface.
 4. When a user selects an option, the corresponding **price adjustments are applied**.
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Updating Door Configuration Options

To update the configuration options:

1. Go to **Shopify Admin**
2. Navigate to **Products**
3. Open the desired product
4. Scroll to the **custom.door_options** metafield
5. Modify the relevant JSON option (for example **Door Configuration**)
6. Save the product

The storefront will automatically reflect the updated configuration.



3. Product Categories & Filtering

Product categories are stored in the **product_category** metafield.

This field stores category assignments in **JSON array format**.

Example JSON

```
[
  "Steel Metal",
  "Exterior Entry Doors",
  "Interior Doors"
]
```

How to Update Categories

1. Go to **Shopify Admin**

2. Open the product
3. Find the **Product Category metafield**
4. Edit the **JSON list**
5. Save the product

The storefront automatically reads this data to:

- Display products in the correct categories
- Enable sidebar filtering
- Organize the catalog

This provides a flexible category system similar to **Magento**, but simpler to maintain.

Door Model	
Zoho Item ID	
R-Value	
OITC Rating	
STC Rating	
SHGC	
U-Factor	
Project Application	
Warranty	
Custom Capabilities	
Impact Rated	
Fire Rated	
Construction	
Wood Species	
Pre-Hanging	
Finish	
Species	

Unpinned

Tab Lead Time

<div id="product_tabs_custom4_contents"> <div class="product-specs"> <p><span style="text-de...

Tab Specification

<div> <ul type="disc"> Pre-Hanging Available: single door and double door o...

Tab Overview

<div style="margin: 0in 0in 0pt;"><span style="color: #ff0...

Product Category

Single line text (List)

Interior & Closet Doors

Shop by Brand

Louvered Doors

Solid Core Doors

4. Product Tab Details

Each product page contains three tabs:

- **Overview**
- **Specification**
- **Lead Time**

These tabs are also controlled through **product metafields**.

Overview

Displays the main product information or summary of the door.

Specification

Displays technical details such as:

- materials
- door dimensions
- glass type compatibility
- other technical attributes

Lead Time

Displays production and shipping timelines for the door.

How to Update Product Tab Content

1. Go to **Shopify Admin**
2. Open the product
3. Locate the metafields for:
 - Overview
 - Specification
 - Lead Time
4. Update the text content
5. Save

The storefront theme will automatically populate the product tabs using this data.

No Liquid files need to be modified.

Unpinned	
Tab Lead Time	<code><div id="product_tabs_custom4_contents"> <div class="product-specs"> <p><span style="text-de...</code>
Tab Specification	<code><div> <ul type="disc"> Pre-Hanging Available: single door and double door o...</code>
Tab Overview	<code><div style="margin: 0in 0in 0pt;"><span style="color: #ff0...</code>

5. Important Note About Data Format

Two metafields require **valid JSON format**:

- **Custom Options**
- **Product Category**

When editing these fields, the structure must remain valid JSON. Incorrect formatting may cause the configurator or filters to stop working properly.

6. Benefits of This Architecture

Easy Maintenance

All updates can be done directly in Shopify Admin.

No Developer Required

Your team can update:

- pricing
- options
- categories
- product information

without editing theme code.

Dynamic System

The theme dynamically reads metafields and builds the interface automatically.

Scalable

New products, options, and categories can be added without modifying the system architecture.

7. Product Import Method

Most product data can be imported using **Shopify's native product importer**. This includes standard fields such as:

- Product Title
- Handle
- Description
- Images
- Variants (including Size)
- SKU
- Price
- Inventory

However, two fields in this system use **JSON formatted metafields**, which the native Shopify importer does not handle well:

- `custom.door_options`
- `product_category`

Because these fields contain structured JSON data, they usually cannot be reliably imported through the default CSV importer.

Recommended Import Approach

The recommended workflow is:

Step 1 — Import Standard Product Data

Use the **Shopify native product CSV importer** to import all standard product fields such as:

- title
- handle
- description
- images

- variants
- pricing
- inventory

This works perfectly for most of the product data.

Step 2 — Update JSON Metafields Manually

After the product is imported, the following metafields can be added or edited directly inside **Shopify Admin**:

- `custom.door_options`
- `product_category`

To update these fields:

1. Go to **Shopify Admin**
 2. Navigate to **Products**
 3. Open the product
 4. Scroll to the **Metafields** section
 5. Paste or edit the **JSON data**
 6. Save the product
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Optional: Using Matrixify for Bulk Imports

If a large number of products need to be imported with JSON metafields already included, a tool such as **Matrixify** can be used.

Matrixify supports importing:

- metafields
- JSON structured data
- bulk product updates

However, using Matrixify is **optional**. For most workflows, it is perfectly acceptable to:

1. Import products using Shopify's native importer
 2. Manually update the JSON metafields afterward.
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Summary

- Standard product fields → import using **Shopify CSV importer**
- JSON metafields (`custom.door_options`, `product_category`) → update **manually in Shopify Admin**
- Matrixify → optional tool for **large-scale bulk metafield imports**

7. Summary

The system has been designed as a **fully dynamic, plug-and-play architecture**.

Your team will be able to manage:

- Door configurations
- Pricing rules
- Categories
- Product information
- Tab content (Overview, Specification, Lead Time)

directly from the **Shopify dashboard**.